In the above code, the body of a for loop enclosed in braces is not executed because a semicolon is given after the for loop.

Declaration of variable in a for loop

In C++, the variables can also be declared within a for loop. For instance,

```
int main ()
{
   int sum = 0;
   for(int i=0;  i<=5;  ++i )
   {
         sum = sum + i;
   }
   cout<<"\nThe variable i cannot be accessed here";
   cout<<"\n The variable sum can be accessed here";
}
```

Variable (i)is declared within the for loop.

The variable i can be accessed only within the body of loop.

A variable declared inside the block of main() can be accessed anywhere inside main() i.e., the scope of variable in main()

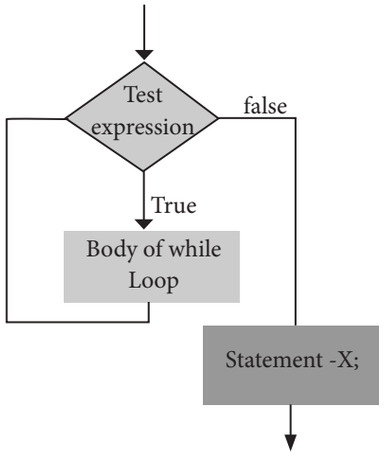
10.5.3 While loop

A while loop is a control flow statement that allows the loop statements to be executed as long as the condition is true. The while loop is an entry-controlled loop because the test-expression is evaluated before entering into a loop.

The while loop syntax is:

```
while ( Test expression )
{
        Body of the loop;
}
Statement-x;
```

The control flow and flow chart of the while loop is shown below.



Flowchart 10.7: while loop control flow and flowchart

11th Std Computer Science_EM Chapter 10.indd 168 06-01-2023 15:09:57